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Systematic Review and Meta-Analysis

Total Hip Arthroplasty in Human Immunodeficiency Virus Positive Patients: A Systematic Review of Outcomes

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ABSTRACT

Background: Total hip arthroplasty (THA) has become a common intervention for human immunodeficiency virus (HIV)-positive patients who have osteonecrosis of the femoral head. This paper provides a systematic review to assess survivorships, patient-reported outcomes (PROMs), infection rates, other complications, and immune competence for patients who had THAs who did and did not have HIV.

Methods: A comprehensive and systematic review of published studies investigating the outcomes of THA in HIV-positive patients (osteonecrosis and non-osteonecrosis patients) was performed adhering to Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines. A total of 10 studies with 9,534 HIV-positive patients fulfilled the inclusion and exclusion criteria. Manuscript quality was assessed on 2 scales: the Coleman Methodology Score and the level of evidence derived from Center for Evidence-Based Medicine criteria. Data were extracted from studies in the 5 key areas of interest: survivorships, PROMs, infection rates, complications, and clusters of differentiation-4 (CD4+) counts and viral loads.

Results: Implant survivorship was between 95% and 100%. Postoperative Harris Hip Scores were significantly improved from preoperative values in HIV-positive patients. Postoperative PROMs and infections did not appear to be different between HIV-positive and HIV-negative patients. Many of the cohort studies demonstrate comparable complication rates to matched controls. Where described (7 reports), mean CD4+ counts ranged from 425 to 646 cells/mm³, with low viral loads (3 reports) and variations not found to influence outcomes.

Conclusions: THA is an effective treatment for HIV-positive patients, many of whom suffer from osteonecrosis of the femoral head. The results demonstrate excellent implant survivorship, improved quality of life, and a low risk of infections and complications.

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Human immunodeficiency virus (HIV) infection was first documented in humans in the early 1980s. As of the end of 2020,

there were an estimated 37.7 million people living with HIV infections worldwide. During the 1980s and 1990s, these infections led to rapid immune system deterioration and dramatically shortened life expectancies for those affected. Following the advent and successful implementation of highly active antiretroviral therapy (HAART) programs in 1997 [1], the annual incidence of new infections decreased. The goal of HAART therapy is to sustain undetectable plasma viremia, which helps prevent transmission rates and limits opportunistic diseases [2]. With the widespread use of these medications, patients who have HIV are approaching a near-normal life expectancy when compared to matched uninfected

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control populations [2]. Thus, this population is now experiencing degenerative joint diseases and undergoing arthroplasty procedures at a much higher rate than in prior decades.

Additionally, osteonecrosis of the femoral head (ONFH) is a well-known and debilitating complication of HIV infection [3]. An increased incidence (2.47 per 1,000 person-years in HIV-infected patients as compared with a population-based incidence of 0.04 per 1,000 person-years) of this disease in HIV-infected patients has been reported [4,5]. The incidence of ONFH in HIV-positive individuals is reported to be between 0.34% and 0.65%, which is 45 to 100 times higher than in the general population [5–7] and affects a relatively young population (mean age 34 years) [8]. The progression of osteonecrosis of the hip often results in symptomatic end-stage arthritis. Many of these patients will therefore require a total hip arthroplasty (THA) at a younger age than patients affected by primary osteoarthritis.

HIV infections lead to progressive deterioration of the immune system and an increased risk of infection. Periprosthetic joint infection (PJI) is the most common indication for revision hip arthroplasties [9,10]. As PJIs have major implications for patients and healthcare systems, it remains clinically relevant to better understand the potential risk for HIV-positive patients. It has been recognized that as the immune system status deteriorates, the HIV-positive patient who has a prosthetic implant may be at risk for later hematogenous implant-related infection [3]. To assess immune competence, investigators have typically monitored patient cluster differentiation-4 counts (CD4+) and viral loads (VLs) to assess immune competence. There is a spectrum of reports in the literature, but few reviews and little consensus on optimal treatment for those patients requiring arthroplasty. Of note, this manuscript will not include patients who have hemophilia or intravenous drug use, as there is already evidence in the literature of higher rates of PJIs and complications in this population [11–14].

This paper provides a systematic review, taking into account levels of evidence using a methodology index, to assess survivorships, patient-reported outcomes, infections, and other complications for those who have THA and HIV (including osteonecrosis and non-osteonecrosis patients). In addition, we will detail CD4+ counts and VLs when described as a reflection of the immune status of the patient populations and whether this has impacted patient outcomes.

Methods

We performed a comprehensive and systematic review of published studies investigating the outcomes of THA in HIV-positive patients, adhering strictly to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines (see Figure 1) [15]. A rigorous search of the PubMed, Cochrane Library, Embase, and Medline databases was carried out up until July 1, 2023.

Selection Criteria

Our search encompassed terms such as “HIV,” “AIDS,” “arthroplasty,” “total hip replacement,” “hip surgery,” “osteonecrosis,” and “ONFH.” The initially discovered reports were examined, and their bibliographies were cross-referenced to find additional relevant literature (Figure 1). The inclusion criteria comprised full-text articles in English, whereas meta-analyses, systematic reviews, conference abstracts, and letters to the editor were excluded. Duplicate entries were also removed.

There were 2 authors (A.B.L. and D.H.) who independently performed the initial search using the selection criteria to identify applicable reports among the specified databases. A third author (M.A.M.) resolved any discrepancies in the primary selection of articles.

A total of 10 studies with 9,534 HIV-positive patients fulfilled the inclusion and exclusion criteria for analyses investigating the outcomes of THA in HIV-positive patients.

Data Syntheses

To evaluate the quality of each clinical study, the Coleman Methodology Score was used (Table 1) [26]. This well-established tool lends credibility to research findings by providing an objective and standardized examination of study designs, data collections, and data analyses. This rating scale measures sample sizes and follow-up times, as well as the effectiveness of clinical measurements, and is often used to compare procedural investigations. A table with specific scoring values used in this study can be found in Appendix 1. As noted, this modified score ranges from 0 to 100, where scores of 85 to 100 are considered “excellent,” 70 to 84 are deemed “good,” 55 to 69 are labeled “fair,” and those scoring less than 55 are regarded as “poor [26].” There were 3 studies that scored between 80 and 85, placing them in the “excellent” category. Seven studies were classified as “good,” with scores between 70 and 79. No studies fell into the “fair” or “poor” category.

Although publication bias was not assessed due to the limited number of studies, the level of evidence was evaluated based on the Oxford Center for Evidence-Based Medicine criteria [27]. These criteria categorize medical research into different levels of evidence, ranging from I to IV. Level I studies represent the highest

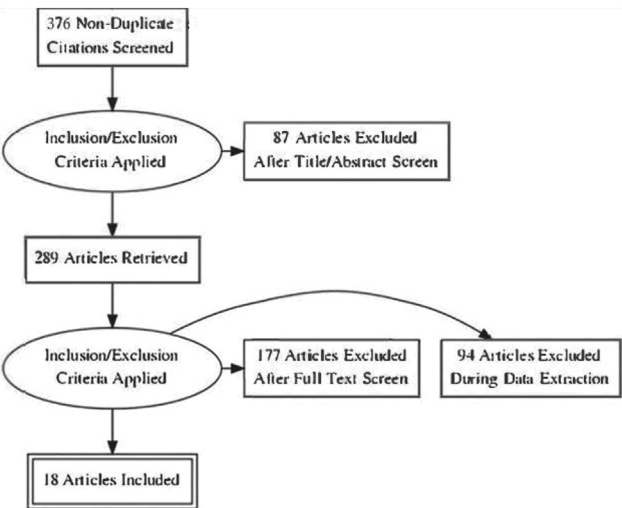


Fig. 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flowchart delineating the study selection process.

Table 1 Modified Coleman Methodology Scores and Level of Evidence.

Study	Score	LOE
Issa et al., 2013 [16]	85	II
Tornero et al., 2012 [17]	81	III
Sax et al., 2021 [18]	80	III
Zhao et al., 2015 [19]	78	III
Graham et al., 2014 [20]	77	III
Naziri et al., 2015 [21]	75	III
Falakassa et al., 2014 [22]	71	III
Baek et al., 2023 [23]	76	IV
Brijlall et al., 2008 [24]	74	IV
Graham et al., 2021 [25]	72	IV

LOE, level of evidence.

quality of evidence, including systematic reviews and randomized controlled trials, while Level IV includes less reliable sources like case reports and expert opinions. In this evaluation, Level II and III studies might encompass well-designed controlled trials and retrospective studies, respectively.

In Table 1, the studies are grouped into levels of evidence, with 1 study in Level II, 6 in Level III, and 3 in Level IV. There were 3 studies that scored between 80 and 85, placing them in the “excellent” category. Seven studies were classified as “good,” with scores between 70 and 79. No studies fell into the “fair” or “poor” category.

Outcomes of Interest

Studies were categorized based on the specific outcomes of interest, leading to a focused subanalysis of 5 key areas: survivorships, patient-reported outcome measures (PROMs), infection rates, complications, and immune competence (CD4+ counts and VLs) [28]. Survivorships were evaluated by examining the long-term functionality and durability of THA as defined by all-cause revision rates. The analyses of PROMs provided insight into patients’ self-reported experiences, encompassing aspects such as pain, mobility, quality of life, and satisfaction with THA. Infection rates (PJIs) as well as other complications were analyzed. To facilitate a comprehensive review, the following data were extracted from each included article where available: author, study design, sample size, demographics, length of follow-up, comorbidities, infection rate, revision rate, and complications.

Assessment of Immune Competence [28]

Clusters of differentiation-4 (CD4+) are glycoproteins present on the surface of white blood cells. The CD4+ T-cells are considered to play a key role in immune protection and are destroyed by the HIV virus. Normal CD4+ counts range from 500 to 1,500 cells/mm³. Recent guidelines allow for the continuation of a surgical plan if the counts are more than 200 cells/mm³. VL is the number of virus particles found in a quantity of 1 milliliter of blood. If the number of particles was less than 50 to 100 copies/mL, in all studies this was considered ‘undetectable,’ and would be the most advantageous time to perform elective surgery.

Data Analyses

Due to the inherent methodological and clinical heterogeneity of the included studies, a narrative review of the results has been presented. Initially, the collected data were segregated based on the outcomes of interest. Following the data categorization, the next step was the extraction process. The database was carefully structured and prepared using a Microsoft Excel spreadsheet (Redmond, Washington) to manage, filter, and interpret the data.

Results

Survivorships

In this cohort, 6 studies reported on survivorship, and overall, there were similarities to HIV-negative survivorship in HIV-positive THAs. Issa et al. reported on 34 HIV-positive patients (44 hips) who underwent primary THA for the treatment of ONFH and were compared with a control cohort of 70 patients (78 hips) who did not have HIV or other high-risk factors for ONFH [16]. Kaplan-Meier survival analysis demonstrated no significant difference in aseptic implant survivorship between the HIV-positive and HIV-negative cohorts at the 5-year (100 versus 98%, respectively) and 10-year (95 versus 96.5%, respectively) follow-up periods [16].

Falakassa et al. reported on 24 HIV-positive patients who underwent 32 primary THAs [22]. One patient needed a revision for mechanical (aseptic) loosening. In the Graham et al. publication on the Southeast African population, there were 43 THAs followed for a mean of 42 months, and none required revision surgeries [20]. However, they continued enrolling patients, and in a cohort of 102 THAs with a mean follow-up of 51 months, there were 6 hips (5.9%) that developed aseptic loosening of the prosthesis. They reported that 5 of these patients underwent revision surgery [25].

Zhao et al. reported a retrospective study of patients who underwent THA for ONFH in Beijing [19]. In their series, 28 HIV-positive patients and 35 HIV-negative patients underwent 48 THAs and 45 THAs with cementless prostheses, respectively. No revision surgery was required in any patient. Baek et al. performed a retrospective study of 22 hips in 15 patients and found no cases of revision in their mean 5.2-year follow-up [23]. Additionally, they performed a radiographic assessment, and there was no evidence of focal osteolysis or loosening of the prosthesis; all implants showed stable fixation by osseous ingrowth. Brijlall et al. studied 15 hips that underwent THA in HIV patients who had similar results—no evidence of loosening of the prosthesis and no cases of revision [24].

Patient-Reported Outcomes

There were 7 studies that described patient-reported outcome data (Table 2). Overall, HIV-positive patients had excellent outcomes compared to non-HIV patients in all studies. Graham et al. reviewed 29 patients (43 THAs) at a mean follow-up of 42 months (range, 5 to 98 months) [20]. There was a substantial improvement in the average Harris Hip Score, from 27 preoperatively to 86 postoperatively, giving a mean improvement of 59 points ($P < .05$). Graham et al. built off this cohort, continued enrolling, and evaluated longer-term outcomes (mean 51 months) [25]. The mean preoperative Oxford Hip Score was 14.0 and improved to a mean of 46.6 postoperatively, an improvement of 32.6 ($P \leq .001$). The mean preoperative Harris Hip Score was 29.4 and improved to a mean of 85.0 postoperatively, an improvement of 55.6 ($P \leq .001$) [25].

In the Issa et al. manuscript discussed previously, with 34 HIV-positive patients undergoing 44 THAs, the mean postoperative Harris Hip Scores (85 points in the HIV group versus 87 points in the comparison group), activity scores (5.7 points in the HIV group versus 6.1 points in the comparison group), SF-36 physical component scores (43 points in the HIV group versus 46 points in the comparison group), and SF-36 mental component summary scores (54 points in the HIV group versus 57 points in the comparison group) were statistically similar between the 2 cohorts [16]. Additionally, Zhao et al. found that in HIV-positive patients, the Harris Hip Score improved from a preoperative mean of 25.2 to a postoperative mean of 82.5, whereas in HIV-negative patients, the Harris Hip Score improved similarly from a preoperative mean of 28.9 to a postoperative mean of 84.3 ($P < .05$) [19].

Other reports had similar positive patient-reported outcomes. Brijlall et al. report on 14 patients who had 15 THAs and were fully ambulatory at a mean follow-up of 72 months [24]. A cohort of 13 HIV-positive patients (18 hips) and 27 HIV-negative patients (36 hips) from Spain demonstrates similar postoperative functional results, which were maintained until the end of the follow-up period (mean 3.3 years in the HIV-positive group and 5.5 years in the HIV-negative group; $P = .08$) [17]. Baek et al. performed a retrospective study of 15 HIV-positive patients (22 hips) who underwent cementless THAs [23]. The mean postoperative follow-up was 5.2 years (range, 1 to 16 years). The average modified Harris Hip Score improved from 58.3 preoperatively to 95.2 at the latest follow-up [23].

Table 2
Outcomes in Total Hip Arthroplasty Patients Who Have HIV.

Category	Study Population	Study Aim	Survivorship	Complications	Infection Rate	Patient-Reported Outcomes
Brijlall et al., 2008 [24]	14 patients (15 hips)	To assess the clinical outcome of THA in HIV-positive patients	100% survivorship with no revisions noted.	(0%) No cases of hip dislocation or infection	0%	All patients were fully ambulatory at a mean follow-up of 72 mo
Tornero et al., 2012 [17]	13 HIV + patients (18 hips) and 27 HIV- patients (36 hips)	To compare outcomes of THA for ONFH between HIV-positive and HIV-negative patients	Not reported	Operative time: HIV+: 106 min, HIV-: 109 min ($P = .66$); Hospitalization: HIV+: 7.8 d, HIV-: 9.4 d ($P = .16$); Blood loss: HIV+: 2.7 g/dL, HIV-: 3 g/dL ($P = .56$); Need for red cell transfusion: HIV+: 1, HIV-: 4 ($P = .48$)	Not reported	Similar postoperative functional results maintained till end of follow-up period
Issa et al., 2013 [16]	34 HIV + patients (44 hips) and 70 HIV- patients (78 hips)	To compare outcomes of THA for ONFH between HIV-positive patients and a control cohort	No significant difference in aseptic implant survivorship between the HIV and comparison cohorts at the 5-y (100% versus 98%, respectively) and 10-y (95% versus 96.5%, respectively) follow-up times	Not reported	4.5% (2 infections in HIV-positive cohort)	Harris Hip Scores: HIV+: 85, HIV-: 87; Activity scores: HIV+: 5.7, HIV-: 6.1; SF-36 physical scores: HIV+: 43, HIV-: 46; SF-36 mental scores: HIV+: 54, HIV-: 57. No significant differences between HIV-positive and HIV-negative cohorts
Falakassa et al., 2014 [22]	24 HIV + patients (32 THAs)	To assess outcomes of THA and TKA in HIV-positive patients	96% survivorship: HIV patient: 1 revision (aseptic loosening) at 1-y postoperative	MSSA Bacteremia: 1/24; Pneumonia/Respiratory Failure: 1/24.	0% PJI	Not reported
Graham et al., 2014 [20]	29 HIV-positive patients	To review outcomes of THA in HIV-positive patients	100% survivorship in early cohort	No early (< 6 wk) or late (> 6 wk) complications such as hip dislocation, superficial or deep wound infection, deep-vein thrombosis (DVT), or pulmonary embolism	0%	Preoperative Harris Hip Score 27; Postoperative Harris Hip Score 86; mean improvement of 59 points ($P < .05$)
Graham et al., 2021 [25]	102 patients (102 THAs)	To assess longer- term outcomes of HIV-positive patients undergoing THA	THA aseptic loosening (6 hips); 5 revisions Survivorship: 95.1%	Only complication reported- Deep infection treated with DAIR	1 deep infection (0.98%)	Preoperative Oxford Hip Score: 14.0; Postoperative Oxford Hip Score: 46.6 ($P \leq .001$); Preoperative Harris Hip Score: 29.4; Postoperative Harris Hip Score: 85.0 ($P \leq .001$)
Naziri et al., 2015 [21]	2,656,696 HIV- patients and 9,275 HIV + patients	To compare outcomes of THA between HIV-positive and HIV-negative patients in the United States	Not reported	HIV+: Acute renal failure (1.6%), pneumonia (0.8%), wound hemorrhage (1.6%), wound infection (0.7%); HIV-: Acute renal failure (0.9%), pneumonia (0.6%), wound hemorrhage (1.1%), wound infection (0.2%); Length of hospital stay: HIV+: 4.31 d, HIV-3.83 d ($P < .001$)	0.7% in HIV-positive patients	Not reported
Zhao et al., 2015 [19]	28 HIV + patients and 35 HIV- patients	To compare outcomes of THA for ONFH between HIV-positive and HIV-negative patients	100% Survivorship; No revision surgery was required in any patient.	Operative time: HIV+: longer ($P < .05$); Blood loss and hospital stay: no difference ($P > .05$)	0%	Harris Hip Score preop: HIV+: 25.2, HIV-: 28.9; Harris Hip Score postop: HIV+: 82.5, HIV-: 84.3 ($P < .05$)
Sax et al., 2021 [18]	Patients who have HIV who were and were not taking ART, and a HIV-negative cohort	To compare outcomes of THA among different patient groups	Not Reported	Length of hospital stay: HIV + on ART: 4.3 d, HIV + not on ART: 4.3 d, HIV-: 4.1 d; Readmissions: HIV + on ART: 4.2%, HIV + not on ART: 3.5%, HIV-: 2.2%;	Periprosthetic joint infection rates at 1 y: HIV + not on ART: 5.3%, HIV + on ART: 4.2%, HIV-: 3.8%	Not reported
Baek et al., 2023 [23]	15 patients (22 hips)	To evaluate outcomes of THA for ONFH in HIV-positive patients	100% Survivorship; No revision surgery was required in any patient.	1 incidence of pneumonia postoperatively	0%	Modified Harris Hip Score improved from 58.3 preoperatively to 95.2 at the latest follow-up

THA, total hip arthroplasty; ONFH, osteonecrosis of the femoral head; ART, antiretroviral therapy; TKA, total knee arthroplasty; HIV, Human Immunodeficiency Virus.

Infections

When assessing the HIV-positive patients, these data support an infection rate between 0.5% and 3%. Although many of these studies had small patient populations and the larger database registries had limited follow-up, this is an encouraging finding. Many of these studies reported 0 infections in their cohorts, including Brijlall et al., Zhao et al., and Baek et al. [19,23,24]. Of note, in the manuscript by Sax et al., they analyzed subsets of patients who were and were not taking HAART [18]. The HIV-positive patients on HAART had infection rates of 4.2%, while those not on medications had an infection rate of 5.3%. Both were slightly higher than the HIV-negative control group, which had an infection rate of 3.8%.

Complications

Most of the 9 cohort studies demonstrate comparable complication rates that were similar to matched controls. In fact, a number of the included studies report no complications in either group, which may be attributable to small sample sizes. Many of these reports are likely underpowered to detect such differences. When a larger database was queried, there were higher complication rates for patients who had HIV and were undergoing THA. Naziri et al. used the National Inpatient Sample to compare HIV-positive and HIV-negative patients who underwent THA between 1998 and 2010 in the United States [21]. The study population consisted of 2,656,696 patients who did not have HIV and 9,275 patients who had HIV. After controlling for confounding variables (including age, sex, race, type of insurance, and medical comorbidities), HIV-positive patients had an increased rate of major complications (2.9 versus 2.7%; $P = .014$), an increased rate of minor complications (5.2 versus 4.8%; $P < .001$), and an increased length of hospital stay (4.31 versus 3.83 days, $P < .001$). Although these differences were statistically significant, these differences are small and appear to be less clinically relevant [21,29].

CD4+ Counts and Viral Loads

There were 7 studies that described mean preoperative CD4+ counts and 3 that noted preoperative VLs that ranged from 425 to 646 cells/mm³. No studies described that any variations influenced outcomes. Brijlall et al. found that the mean CD4+ count was 425 cells/mm³ [24]. Tornero et al. found the most recent VL within the 3 months prior to surgery was < 50 copies/mL in all patients, except for 1 case [17]. The most recent CD4+ count within the 3-month period prior to surgical intervention was a mean of 434 cells/mm³. Falakassa et al. reported preoperative mean CD4+ counts for the HIV-positive patients of 646 cells/mm³ [22]. The mean VL at the time of surgery was undetectable in 19 patients, and 3 patients had detectable VLs of < 100 copies/mL. Patients in the first study by Graham et al. had an average preoperative CD4+ count of 489 cells/mm³ [20]. Zhao et al. reported a mean preoperative CD4+ count of 393 cells/mm³ in their patients [19]. Of 28 HIV-positive patients, the VL was undetectable in 20. Graham et al. reported a mean CD4+ count of 529 cells/mm³ [25]. The patients in Baek et al. had an average preoperative CD4+ count of 476 cells/mm³ [23]. Preoperative HIV VL was negative (undetectable) in 15 cases (68.2%), < 200 copies/mL in 4 cases (18.2%), and > 200 copies/mL in 3 cases (13.6%).

Discussion

THA has become a feasible intervention for addressing ONFH in HIV-positive patients. As advancements in medical care continue to enhance the life expectancy of this demographic, the relevance of

THA has grown proportionally. This systematic review explores the multifaceted outcomes of THA within this specific patient group, elucidating results on implant survivorships, patient-reported outcomes, infection rates, and complications.

One of the findings from the data is the longevity and success rate of THA implants in HIV-positive patients. Contrary to initial concerns, these implants do not exhibit a higher failure rate in the HIV-positive community compared to the general population. Another highlight from the review is the marked improvement in the Harris Hip scores among HIV patients post-THA. Their feedback and satisfaction levels were comparable to those of primary osteoarthritis patients who underwent the same procedure. This not only underscores the efficacy of THA for treating ONFH but also emphasizes its transformative impact on patient mobility and quality of life.

Furthermore, the review makes an intriguing observation regarding infection rates. HIV-positive patients, especially those who do not have coexisting conditions like hemophilia or intravenous drug use, exhibited surprisingly low post-THA infection rates. This is noteworthy since one might expect higher infection rates in immunocompromised individuals. It suggests that with effective antiretroviral therapy, such as HAART, and well-managed immune function, HIV-positive patients can undergo THA with an infection risk that may be slightly higher, which aligns closely with the general population.

These excellent results for HIV+ patients who had THAs are likely due to the modern approach to their health. In most studies, there was a description of the assessment and optimization of the patient's immune status. This can be exemplified by the fact that in 7 of the studies, mean CD4+ counts were more than a mean of 393, and in 3 studies, there were low or undetectable VLs reported when patients were operated upon.

These patients should have preoperative assessment and optimization. The HIV-positive patients may have associated medical comorbidities such as coronary artery disease, hepatic dysfunction, renal disease, malnutrition, coagulopathy, pancytopenia, substance abuse, a history of opportunistic infections, neurological complications, and malignant conditions [30,31]. Therefore, a detailed preoperative evaluation of HIV-positive patients is vital before undertaking total joint arthroplasty. Evaluation of nutritional status is especially important; hypoalbuminemia has been reported as an independent risk factor for postoperative mortality in HIV-positive patients when compared to uninfected controls [32]. Malnutrition is associated with several complications after THA, including delayed wound healing, persistent wound drainage with subsequent susceptibility to infection, and prolonged hospital stays. Likewise, these patients need to have a thorough evaluation of their cardio-pulmonary status before undergoing these procedures.

We also believe that the CD4+ cell counts should be followed postoperatively by the infectious disease specialist and should also be reviewed at regular follow-up visits with the orthopedic surgeon given that there have been some suggestions that complications may be associated with declining CD4+ counts because of either worsening of the HIV disease process or diminished adherence to HAART [33].

Advantages of this systematic review are that it is a complete overview of all studies on the topic to date, with an analysis separated by survivorships, PROMs, infection rates, and complications. Unfortunately, the challenge with this systematic review lies in the inherent limitations of the available research. Many studies suffer from small sample sizes, limited follow-up durations, and retrospective designs, leading to potential biases and gaps in the data. Of the 10 studies included, one paper had a disproportionate number of patients. Nevertheless, the results were consistent for the remaining papers in regards to the outcome parameters

(survivorships, complications, infection rates, and patient-reported outcomes). The survivorship data were based on small studies because of the available literature, with a need for more long-term studies.

Although ONFH is common in the HIV population, this paper did not focus on this group. In fact, many patients in both groups were not ONFHs. Of note is that this is a comparison of total hips in HIV and non-HIV patients and not specifically looking at ONFH, as the information on ONFH subcategorization was not available in the studies. One should note that survivorships were based on only 6 studies with a limited number of patients each. We think that all of these 6 studies were consistent with excellent results, so one can be optimistic to date. Nevertheless, we agree that the small number of total patients for outcomes is a limitation of the study that we cannot overcome, since this is the world literature to date on this topic. We await larger studies in the future. Also, it would have been nice to know the CD4+ level in all patients who underwent THAs. Those patients who had lower CD4+ levels might show worse outcomes, for example, increased complication rates, including surgical site infection and PJI, as well as longer hospital stays.

Conclusions

The insights derived from this discussion reiterate that THA is an effective treatment for HIV-positive patients suffering from ONFH. The results demonstrate the potential to improve quality of life with a reasonable risk profile. The ongoing evolution of medical management with the use of HAART therapy and the extension of life expectancy for this population make this research both timely and vital, underscoring the importance of continued exploration and refinement in this area. There were slightly increased infection rates found, but these were not high enough to decrease the excellent PROM scores, low complication rates, and survivorships in this patient population.

CRediT authorship contribution statement

Alexandra Baker Lutz: Writing – original draft. **Deepak V. Patel:** Writing – review & editing. **Christopher Bibbo:** Writing – review & editing, Conceptualization. **Ian S. Hong:** Writing – review & editing. **Daniel Hameed:** Writing – review & editing. **Jeremy Dubin:** Writing – review & editing. **Michael A. Mont:** Writing – review & editing, Conceptualization.

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Appendix

Supplementary Table 1

Modified Coleman Methodology Score.

Part A: Only <i>one</i> Score to be Given for Each of the 7 Sections	
1. Study size: number of patients (total)	
<30	0
30-50	4
51-100	7
>100	10
2. Mean study duration	
<12 mo	0
12-36 mo	4
37-60 mo	7
>60 mo	10
3. Surgical approach	
Different approach used and outcome not reported separately	0
Different approaches used and outcome reported separately	7
Single approach used	10
4. Type of study	
Retrospective cohort study	0
Prospective cohort study	10
Randomized controlled trial	15
5. Description of diagnoses	
Described without percentage specified	0
Described with percentage specified	5
6. Descriptions of surgical technique	
Inadequate (not stated, unclear)	0
Fair (technique only stated)	5
Adequate (technique stated, details of injection given)	10
7. Description of postoperative rehabilitation	
Described	5
Not described	0
Part B: Scores may be given for each option in each of the 3 sections	
1. Outcome criteria	
Outcome measures clearly defined	2
Timing of outcome assessment clearly stated	2
Use of outcome criteria that has reported reliability	3
General health measure included	3
2. Procedure of assessing outcomes	
Participants recruited	5
Investigator independent of surgeon	4
Written assessment	3
Completion of assessment by patients	3
3. Description of subject selection process	
Selection criteria reported and unbiased	5
Recruitment rate reported	
>90%	5
<90%	0
Total Part A	/65
Total Part B	/35
Total	/100